Overcoming outcome-label scarcity through population pretraining and adaptive sharing across haematological diseases

[Authors and affiliations]

Working manuscript — October 9, 2026

Abstract

Clinical decisions in haematology depend on predicting treatment benefit, relapse, toxicity and survival, yet the relevant events can be scarce within an individual disease–outcome pair. Population-scale pretraining offers reusable representations from broad patient histories; it does not by itself establish which labelled diseases and outcomes should inform a scarce target. We test these complementary forms of transfer in a nationwide linked haematology resource, we defined 24 fine disease cohorts, 69 approved outcomes and six prediction horizons. We designed a matched readout ladder: pooled multi-horizon binary cross-entropy with a shared representation; disease-source or outcome-source adapters alone; and both source banks with additive or interactive routing. All arms use the same supervised panel, allowing source-bank effects to be distinguished from the effect of pooling training data. A separate encoder ladder tests local pretraining, general-population PHAIR pretraining and haematology domain adaptation under matched downstream readouts. **[Insert locked held-out headline effects, uncertainty, rare-task interaction and XGBoost comparison after the analysis is frozen.]** Together these experiments ask which sources of transferable information actually reduce label scarcity without harmful transfer.

Editorial summary *Population pretraining and cross-task supervised learning offer distinct ways to address sparse disease–outcome labels. OPERA tests whether either improves prediction where events are rare, and whether adaptive routing can retain useful sharing without harmful transfer.*

Introduction

Prediction informs treatment intensity, supportive care, surveillance and trial design in haematology. The difficulty is not merely that some diseases are rare. Once patients are divided by diagnosis, target outcome and prediction horizon, even a large linked cohort can contain few relevant events for an individual clinical question. Censoring and variable follow-up further limit what can be learned and evaluated. This fragmentation of labels is a general problem for specialized clinical prediction, not a peculiarity of haematology. Traditional disease-specific scores, such as the NCCN International Prognostic Index and CLL-IPI, compress a diagnostic snapshot into a transparent rule [1, 2]. Task-specific machine learning can use richer longitudinal histories, but usually aggregates them into fixed windows and fits a separate predictor for each clinical target [3, 4] (Fig. 1). Neither approach automatically solves the small-label problem in a rare cell.

Population-scale pretraining is one proposed response. Sequence models learn from ordered patient histories before specialist outcome labels are supplied [5–7]; broad pretraining has shown promise for label efficiency, including in time-to-event modelling [8]. Its attraction is clear: a model trained on many patients may carry useful structure into a disease with few labelled events. But pretraining does not create target-event labels, and a general patient representation need not be the best representation for every disease and endpoint. Few-shot benchmarks and critical reviews underscore the need to measure transfer on the particular tasks and populations where it is claimed [9, 10]. The relevant question is therefore not simply whether a large pretrained model predicts well, but when its broad experience adds value beyond local pretraining and strong task-specific models.

A second source of information is already labelled: neighbouring diseases and outcomes in the same care system. Multi-task and meta-learning studies show that related prediction tasks can improve low-resource targets [11, 12]. Yet pooling labels need not help every target. Shared supervision may transfer useful risk structure, irrelevant structure, or both; disease and outcome relatedness cannot be assumed from clinical labels alone. This motivates two separable tests: whether joint training beats a specialist fitted to each supported disease–outcome pair, and whether patient-dependent routing improves on a conventional pooled model by selecting which disease and outcome sources to borrow from.

Our earlier gradient-boosted study in diffuse large B-cell lymphoma found that including related aggressive lymphoma subtypes did not degrade prediction of treatment failure, while using fixed lookback windows discarded the order of clinical events [4]. That result concerned one endpoint in closely related diseases; it neither established a general pooling benefit nor tested whether other outcomes can supply useful supervision. We therefore consider not only patients, variables and longitudinal observations, but also the breadth of disease cohorts and outcome tasks available for learning (Fig. 2). Greater breadth creates opportunities for trans-

fer, not a guarantee of improvement.

The central question is whether borrowing either unlabelled patient-history structure or labelled disease–outcome experience improves prediction as target events become scarce. We first compare pooled training against same-encoder specialists across supported disease–outcome pairs; we then test whether disease- and outcome-source routing improves on ordinary pooled learning. A separate, matched encoder comparison asks whether general-population PHAIR pretraining and local domain adaptation add to, or substitute for, supervised sharing. Performance and failures of transfer are examined across the full approved panel rather than inferred from a few favourable endpoints.

To answer these questions we used DALY-CARE, a nationwide linked resource that integrates ten Danish national registers with hospital electronic health records, laboratory results, and disease-specific quality registries for adults diagnosed with lymphoid cancers since 2002 [13]. The resource description reported the cohort available at the time of that report; the present study draws a later extract, so the denominators here are not expected to match those figures. Our study population is also defined differently from our earlier lymphoma analysis. That study indexed at the start of first-line therapy in aggressive lymphoma and asked one question at one horizon, whereas the present study indexes first-line treatment episodes across the full haematological spectrum and evaluates many endpoints at several horizons. The two populations overlap substantially and share a common data substrate, but they are not identical and the results are not directly comparable patient for patient.

On this basis we defined a temporal evaluation with patients indexed from 2023 onward held out, 69 approved outcomes across 24 fine disease cohorts, and six horizons. The primary neural comparison is a prespecified five-arm sharing ladder: pooled shared multi-horizon BCE, disease-source routing only, outcome-source routing only, additive routing over both banks, and interactive routing over both banks. The source-bank contrasts isolate adaptive borrowing *conditional on pooled training*; disease-specific versus joint training is a separate contrast. Clinical scores and task-specific XGBoost remain external comparators once matched feature matrices are locked. The study is also an empirical atlas of what can be predicted from longitudinal records.

Results

A longitudinal prediction atlas across haematological disease

The sealed first-line source contained 37,557 patients, of whom 37,531 had released overall-survival membership and a supervised split before outcome-specific eligibility. Patients were assigned to 24 fine disease cohorts nested within 10 broader disease groups. Cohort size ranged from **[minimum]** to **[maximum]**, creating a natural test of performance under data scarcity. The approved primary panel contains 69 outcomes covering disease control, treatment trajectory, infection, organ toxicity, haematological toxicity, acute care, subsequent malignancy, and survival. Each outcome was evaluated at 30, 60, 90, 180, 365, and 730 days when adequate events and follow-up were available.

[Insert cohort flow and outcome-support results.] The final predictability atlas will report held-out cumulative/dynamic area under the ROC curve (AUC) at 365 days, with all horizons reported in supplementary figures. Calibration and Brier scores will accompany discrimination throughout. Outcomes with inadequate event support are shown as not estimable rather than silently removed.

What is gained by foundation-model pretraining?

The encoder-transfer analysis will compare locally pretrained Time2Vec, general-population PHAIR initialization without local continued pretraining, and PHAIR followed by haematology domain-adaptive pretraining, with the same approved outcome panel and downstream readout. A randomly initialized or local-only encoder serves as a capacity-matched reference where feasible. This ladder separates pretraining scope from local domain adaptation; it is not yet a completed PHAIR result. **[Insert verified PHAIR checkpoint compatibility, matched runs, paired held-out effects and uncertainty.]**

The primary contrast is general-population versus local haematology pretraining under an otherwise matched downstream protocol. The same encoder contrast should be examined under both the shared BCE readout and the selected router: this tests whether better pretraining replaces, complements, or changes the benefit of supervised adaptive sharing. We additionally test whether any advantage is concentrated among patients who are atypical in the local embedding distribution. To assess whether broad pretraining changes the value of labelled-data pooling itself, pooled shared BCE and disease–outcome specialists must also be compared with the same PHAIR-derived encoder on both sides. This is a separate pretraining-by-pooling interaction from the pretraining-by-router interaction; the PHAIR specialist grid follows the same support and evaluation rules as the local-encoder grid.

Because the case for foundation models rests largely on their behaviour when labels are scarce, we do not report this contrast only as a single average. Each pretraining comparison is additionally summarized as a function of cohort size and event support, so that a benefit concentrated in well-populated cohorts is distinguishable from one that materializes where task-specific training is least viable.

Can related cohorts compensate for scarce disease-specific data?

For a full-panel claim about the benefit of joint training, the counterfactual is a specialist trained on each estimable fine-disease–outcome cell, initialized from the same encoder and evaluated against the joint model on exactly the same patients. One specialist can predict

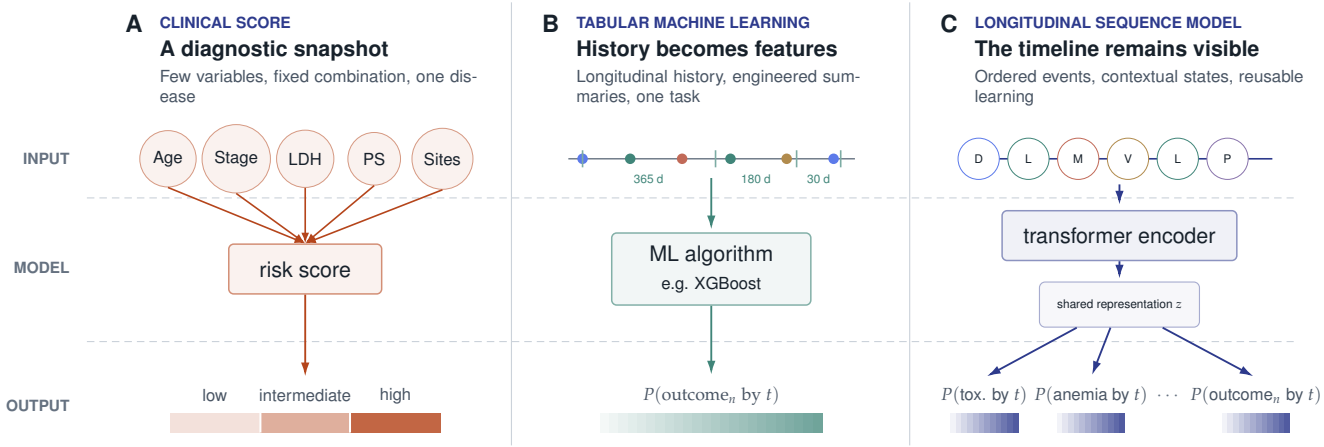


Figure 1. Three paradigms for haematological risk prediction. A, Representative clinical risk scores combine a small diagnostic snapshot using a fixed rule and assign broad risk strata. **B**, classical machine learning uses a wider history but ordinarily requires temporal aggregation into a task-specific feature table. **C**, a longitudinal sequence model consumes ordered clinical events and can share a contextual patient representation across outcomes. Its encoder may be pretrained locally or at foundation-model scale. The comparison concerns information available to each predictor; it does not imply that every score or tabular implementation has exactly the illustrated structure.

all six horizons, so separate models for each horizon are unnecessary. Prespecified event-support thresholds determine which cells can sustain a specialist; unsupported cells remain visible as not estimable. A sampled subset would answer a narrower question and cannot justify a full-panel claim. We model paired performance differences against training event support with partial pooling over diseases, outcomes and horizons. **[Run and report the specialist grid, initialization and budget, posterior joint-training effect, rare-cell interaction and failures of transfer.]**

The main scarcity forest will show paired held-out differences, not absolute AUCs alone. Its principal rows are pooled shared BCE minus the matched disease–outcome specialist (benefit of pooling), interactive routing minus pooled shared BCE (benefit of adaptive borrowing), and interactive routing minus the specialist (total practical gain). Cells without an estimable specialist are displayed separately as a coverage result, never assigned an invented specialist contrast. A companion encoder-transfer panel compares general-population pretraining and local domain adaptation under a fixed readout; it is not mixed into the router contrast. The full outcome-by-outcome forest belongs in the supplement, with the main figure emphasizing the support gradient, clinical-family heterogeneity and lower tail of harm. Within the routed-model panel, an equal-scale additive router is the explicit mechanistic comparator for the interactive router. Their difference is shown separately from the three scarcity contrasts, so interaction is not conflated with either pooling or residual-scale capacity.

This analysis distinguishes population-level performance from task-balanced performance. The former weights patients according to their frequency in the registry; the latter gives each estimable cohort–outcome target equal standing and is central to the rare-disease claim.

Which labelled sources should a patient borrow from?

The main supervised comparison contains five matched readouts: pooled shared multi-horizon BCE; a router with disease-indexed adapter sources only; a router with outcome-indexed sources only; an additive router with both source banks; and an interactive router with both banks (Figs. 7 and 8). Every arm trains on the same pooled patient and outcome panel. Here "disease-only" and "outcome-only" refer to eligible *adapter banks*, not to isolated training cohorts or outcomes. The no-source router is a matched architectural control, distinct from plain shared BCE. Family trunks, outcome attention, factorized readouts and other exploratory architectures are retained as supporting screens rather than additional co-primary hypotheses (Fig. 6).

An earlier architecture screen used a common curriculum and hierarchical support-weighted loss. The approved-panel router comparison uses endpoint-group macro multi-horizon BCE and endpoint-group macro tuning AUC for checkpoint selection. **[Insert verified completed-run roster, common-epoch and best-checkpoint tuning summaries, paired fixed-checkpoint AUC/Brier/calibration contrasts, source-removal effects and uncertainty.]** Because architecture changes may redistribute rather than uniformly improve performance, the confirmatory comparison uses paired predictions and reports the mean effect, the proportion of tasks improved, lower-tail degradation, calibration, and heterogeneity across clinical domains. Outcome reliability weighting and empirically derived task groupings are treated as explicit ablations rather than embedded in the definition of the primary metric.

To separate the two proposed channels of adaptive transfer, the nested 2×2 source-bank ablation (Fig. 8) compares neither bank, disease sources only, outcome sources only, and both banks under the same interactive query. It is complemented by an additive-versus-interactive contrast with both banks. The encoder, pooled data, BCE objective, selection rule and residual



Figure 2. Cohort breadth and task breadth as additional axes of predictive information. **A**, conventional scaling increases the number of patients, variables, or observations available to an isolated prediction problem. **B**, joint modelling can additionally draw on related disease cohorts (rows) and outcomes (columns), although greater breadth need not improve every target. Sharing is restricted to training data and acts through a common encoder and structured readout; it does not transfer future information or assume that diseases and endpoints are exchangeable.

scale must be matched within each contrast; otherwise a scale effect could be mistaken for a query-interaction effect. These contrasts estimate the added value of *source banks* conditional on joint training, not the total effect of pooling diseases or outcomes. The observed 0.1–0.3 residual-scale sweep is reported separately as a capacity sensitivity analysis, with a fresh, code-verified 0.1 reference rather than the historical checkpoint whose interaction branch was absent.

Sequence models versus task-specific tabular learning

For each estimable fine cohort, outcome and horizon, the planned tabular comparison fits XGBoost to locked pre-index EHR feature matrices and evaluates it on the same held-out patients as OPERA. Registry-augmented models and established clinical scores require separate, explicitly reported complete-case populations. The feature-matrix and risk-set alignment are still being audited; no OPERA–XGBoost superiority claim follows from the neural tuning screen. **[Insert verified matched tabular predictions, hierarchical OPERA–XGBoost contrast and forest plot.]**

The purpose of this comparison is not to require one model class to dominate every task. Instead, we estimate where temporal representation, cross-cohort sharing, or cross-outcome sharing changes the performance frontier, and where a task-specific tabular model remains the appropriate benchmark.

The geometry and limits of learned patient representations

We characterize the embedding space through vocabulary neighborhoods, contextual perturbations, patient-level projections, and outcome-density maps. **[Insert representation findings.]** To avoid interpreting a two-dimensional projection as a predictive model, density maps are accompanied by held-out predictive metrics and uncertainty, and colours are defined once across all outcome panels.

Discussion

This study evaluates a proposition that extends beyond a single architecture: clinical information can be increased not only by observing more patients, variables, or time points, but also by structuring how related diseases and outcomes contribute to learning. The linked haematology setting is unusually suited to this question because it combines shared care pathways with marked variation in disease frequency, treatment, follow-up, and endpoint support.

The underlying transfer principle is not new. Multi-task and meta-learning approaches established that related, better-resourced labelled tasks can help a low-resource target [11, 12]. Population-scale self-supervised pretraining draws on a different resource: many patient histories without the target labels. Neither form of transfer is guaranteed to resolve a rare target's label scarcity, and their contributions must be measured separately. Our design does so where scarcity is structural rather than simulated: cohort sizes and event counts vary by orders

of magnitude within a single linked resource and a single care system, so the low-resource cells are real clinical populations rather than subsampled ones.

[Insert concise interpretation of the four principal results.] Several outcomes may remain better served by task-specific gradient boosting, particularly when a compact set of recent measurements dominates risk. Such a finding would not invalidate the foundation-model approach; it identifies the conditions under which representation learning adds value and prevents model complexity from being treated as an end in itself.

The planned tabular comparison is not intended to use a convenient baseline. Its feature construction follows our earlier lymphoma models, which improved on established prognostic indices in that setting [4]. Only after the current pre-index matrix, eligibility and patient matching are audited can we ask whether temporal resolution and structured sharing add information beyond strong task-specific tabular learning. Our earlier observation that training across aggressive lymphoma subtypes did not degrade diffuse large B-cell lymphoma performance motivates a separate, prespecified joint-versus-disease-specific training contrast across the approved panel; the source-bank ablation does not answer that question by itself.

The study also differs from existing clinical foundation-model work in what it varies. Prior time-to-event pretraining has demonstrated that many-outcome objectives improve label efficiency [8], and few-shot benchmarks have been built to measure adaptation under label scarcity [9]. We first hold the encoder fixed while varying the structure of supervised borrowing, then hold the selected readout fixed while varying pretraining scope. This separates two sides of the same transfer problem: learning a broadly useful prior without specialist labels and using related labelled tasks without forcing them to be exchangeable. Reporting the full outcome panel, including cells that are not estimable, is intended to make both benefits and negative transfer visible.

The study also separates three often conflated decisions: how outcomes share model capacity, how noisy tasks are weighted during optimization, and how model performance is averaged for selection. Family definitions used for clinical reporting need not determine loss weights or neural trunks. Our confirmatory analysis therefore treats task reliability and task grouping as explicit, auditable components.

Limitations include the single national healthcare context, residual changes in coding and measurement practices, variation in registry coverage, and the possibility that temporal transport from 2021 to 2023+ reflects both disease and healthcare-system drift. Because the cohort definition, index date, and data extract differ from our earlier lymphoma analysis, the two studies cannot be compared directly, and any apparent change in performance for overlapping endpoints reflects these design differences as well as the change in model class. The large endpoint panel also creates multiplicity and correlated estimands. We address these issues using prospective temporal splits, outcome-specific eligibility, paired patient-level comparisons, hierarchical uncertainty models, and complete reporting of the prediction atlas rather than selective endpoint publication.

Methods

Data sources and study population

All data were obtained from DALY-CARE, which links ten Danish national registers with hospital electronic health records, laboratory results, pathology, microbiology, and disease-specific quality registries for adults diagnosed with lymphoid cancers since 2002 [13]. The resource publication describes the data available at the time of that report; the present study uses a later extract with additional accrued follow-up, and cohort counts therefore differ from the published resource description. **[Insert approved descriptions of the specific linked EHR sources, RKKP/quality data, extract date, ethical approvals, data governance, and linkage keys.]**

The index date is the start of a reconstructed first-line treatment episode. This differs from our earlier lymphoma study, which indexed first-line therapy in aggressive lymphoma only and required registry-confirmed treatment [4]; the present cohort spans all included haematological diseases and admits episodes reconstructed from treatment records where registry confirmation is unavailable (Fig. 3). Patient overlap between the two studies is therefore substantial but partial, and no result reported here should be read as a direct replication of that analysis.

Patients entered a supervised prediction cohort only when registry eligibility and the outcome-specific risk-set definition were satisfied at index time. Self-supervised pretraining could additionally include patients with eligible pre-index longitudinal records, and its denominator is reported separately.

Temporal study design

All splits were defined at the patient level using a shared index date. Training included index dates through 31 December 2021, model selection used index dates from 1 January through 31 December 2022, and held-out evaluation began on 1 January 2023. Held-out labels were not used for architecture, checkpoint, task weight, or hyperparameter selection. Patient histories were truncated at the prediction time. During masked-language-model pretraining, long histories were expanded into non-overlapping context windows so that earlier history remained available without future leakage.

Cohorts, outcomes, and competing events

The canonical inventory contained 24 fine diagnoses nested within 10 grouped cohorts. Outcomes were represented as time to primary event, time to competing death where applicable, or censoring. Outcomes already containing death as part of their definition were modelled as one-cause endpoints. Prediction was evaluated at 30, 60, 90, 180, 365, and 730 days. Full definitions, code lists, eligibility rules, and cohort-flow counts are

provided in **[Supplementary Tables]**.

Sequence representation and pretraining

Longitudinal records were converted to ordered sequences of diagnoses, procedures, medications, laboratory concepts, encounter descriptors, and background tokens. Numerical measurements were represented using prespecified bins in the local sequence runs. The local encoder used a four-layer transformer with a maximum context of 8,192 tokens; Fourier and scaled Time2Vec temporal encodings were compared before the main router screen, which uses the Time2Vec checkpoint. The proposed PHAIR transfer analysis begins from a separately pretrained general-population encoder and either fine-tunes it directly or continues self-supervised pretraining on local haematology records before supervised adaptation. PHAIR checkpoint loading requires a verified mapping of vocabulary, numerical representation, temporal encoding, model dimensions and pretraining history; local Time2Vec pretraining is not itself PHAIR transfer. **[Insert delivered PHAIR encoder specification, compatibility audit, pretraining population and domain-adaptation schedule.]**

Multi-outcome prediction

The direct model produced outcome- and interval-specific primary-event and competing-event hazards, following the discrete-time and competing-risk formulations established for neural survival models [14, 15] and extended here to many outcomes in a single pass [8]. Piecewise-constant hazards were transformed into monotone cumulative incidence functions at each evaluation horizon. We compared exact-time competing-risk likelihood with a censoring-aware multi-horizon binary cross-entropy objective. Where outcomes were weighted by learned task uncertainty, we used homoscedastic uncertainty weighting [16]. A curriculum introduced tasks according to outcome support. Alternative readouts are summarized in Fig. 6.

The primary readout ladder holds the training population, approved outcomes, time representation, loss and selection metric fixed. Disease-only and outcome-only variants mask one adapter source bank but still train jointly on all patients and outcomes. Additive routing sums patient, disease and target queries before scoring sources; interactive routing adds a learned nonlinear function of their combination. Both return a target-specific residual update to the shared patient representation. The residual scale is compared in a separate sensitivity sweep. For a query-mode contrast, scale and all other hyperparameters must match.

Specialist fine-tuning

The primary pooling counterfactual trains one disease–outcome specialist for each estimable cell from the same pretrained encoder as the joint model, with all horizons predicted together. A second specialist arm may start from the pooled supervised checkpoint before cell-specific fine-tuning. This separates the value of joint training as a final predictor from its value as an initialization. A coarser factorial can additionally train one model per disease across outcomes and one per outcome across diseases, isolating the two axes of training-data pooling. These are training-regime comparisons, distinct from adapter source-bank masking within a jointly trained router.

Comparator models

Clinical scores will be evaluated without refitting where their required inputs are available. The XGBoost comparator will be trained on locked EHR-derived feature matrices for the approved cohort–outcome–horizon cells. The intended feature-construction pipeline follows our earlier lymphoma study, aggregating each modality over fixed pre-index lookback windows [4], but the current matrix and eligibility audit remains open. Registry-augmented models will be restricted to patients with required quality variables observed by prediction time, and complete-case support will be reported. Hyperparameters are selected on 2022 tuning data; final model comparisons must use identical held-out subjects and risk sets.

Model selection and statistical analysis

We distinguish checkpoint selection, architecture selection, and held-out reporting. Checkpoints were selected using tuning data only. The confirmatory architecture score first averages horizons within an outcome and then averages outcomes, preventing outcomes with more estimable horizons from receiving greater weight. Population-weighted and equal-cohort summaries are reported as complementary estimands.

Held-out model differences are calculated from paired patient predictions. Bootstrap intervals resample patients within the evaluation population. A hierarchical Bayesian model estimates the overall model contrast and variation by cohort, outcome, horizon, clinical domain, and training support. We report posterior probabilities of superiority and practical equivalence using a prespecified AUC region of practical equivalence **[lock threshold, proposed** ± 0.005**]**. Discrimination is reported using cumulative/dynamic AUC; probabilistic accuracy and calibration are reported using Brier score, calibration intercept, and calibration slope.

Reproducibility

Every run records the temporal split, source checkpoint, configuration hash, seed, cohort and outcome inventory, and software revision. Compact patient- level prediction artifacts are retained for paired comparisons and local figure generation. **[Insert code/data availability statement.]**

Data availability

[Controlled-access statement and application procedure.]

Code availability

[Repository, archival DOI, license, and reproducible environment.]

Acknowledgements

[Funding, collaborators, and compute resources.]

Author contributions

[CRediT statement.]

Competing interests

[Statement.]

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Table 1. Baseline characteristics of the supervised first-line haematology analysis cohort. **[All characteristics except patient counts require re-export from the current release.]**

Characteristic	Overall	Training	Tuning	Held-out
Patients, *n*	37,531	29,105	1,980	6,446
Age, years, median (IQR)	69.3 (59.7–77.1)	68.7 (59.2–76.5)	71.1 (61.1–78.4)	71.7 (61.7–78.8)
Age ≥70 years, *n* (%)	18,011 (48.1%)	13,431 (46.1%)	1,050 (53.4%)	3,530 (55.2%)
Female, *n* (%)	15,647 (41.8%)	12,199 (41.9%)	802 (40.8%)	2,646 (41.4%)
Male, *n* (%)	21,820 (58.2%)	16,909 (58.1%)	1,165 (59.2%)	3,746 (58.6%)
Year of first-line treatment, median (IQR)	2016 (2010–2021)	2014 (2009–2018)	2022 (2022–2022)	2024 (2023–2025)
Observed follow-up, years, median (IQR)	4.1 (1.4–8.6)	5.8 (2.1–10.1)	3.5 (3.2–3.8)	1.3 (0.6–2.2)
Disease groups, *n* (%)				
Amyloidosis	47 (0.1%)	22 (0.1%)	7 (0.4%)	18 (0.3%)
Burkitt/lymphoblastic lymphoma	1,364 (3.6%)	1,075 (3.7%)	87 (4.4%)	202 (3.2%)
Burkitt lymphoma (BL)	295 (0.8%)	238 (0.8%)	15 (0.8%)	42 (0.7%)
Lymphoblastic lymphoma (LBL)	1,069 (2.9%)	837 (2.9%)	72 (3.7%)	160 (2.5%)
CLL/SLL	5,152 (13.8%)	4,092 (14.1%)	244 (12.4%)	816 (12.8%)
Chronic lymphocytic leukaemia (CLL)	4,062 (10.8%)	3,220 (11.1%)	187 (9.5%)	655 (10.2%)
Small lymphocytic lymphoma (SLL)	1,090 (2.9%)	872 (3.0%)	57 (2.9%)	161 (2.5%)
DLBCL and related aggressive B-cell lymphomas	10,408 (27.8%)	8,369 (28.8%)	505 (25.7%)	1,534 (24.0%)
Diffuse large B-cell lymphoma (DLBCL)	9,951 (26.6%)	8,008 (27.5%)	481 (24.5%)	1,462 (22.9%)
Richter transformation (RT)	192 (0.5%)	151 (0.5%)	14 (0.7%)	27 (0.4%)
Transformed follicular lymphoma	265 (0.7%)	210 (0.7%)	10 (0.5%)	45 (0.7%)
Hairy-cell leukaemia	418 (1.1%)	293 (1.0%)	29 (1.5%)	96 (1.5%)
Hodgkin lymphoma	2,644 (7.1%)	2,080 (7.1%)	141 (7.2%)	423 (6.6%)
Indolent B-cell non-Hodgkin lymphoma	5,399 (14.4%)	3,895 (13.4%)	302 (15.4%)	1,202 (18.8%)
Follicular lymphoma (FL)	2,932 (7.8%)	2,248 (7.7%)	150 (7.6%)	534 (8.4%)
B-cell lymphoma, unspecified (BCL)	227 (0.6%)	87 (0.3%)	17 (0.9%)	123 (1.9%)
Lymphoplasmacytic lymphoma (LPL)	1,570 (4.2%)	1,165 (4.0%)	100 (5.1%)	305 (4.8%)
Extranodal marginal-zone lymphoma (EMZL)	500 (1.3%)	318 (1.1%)	33 (1.7%)	149 (2.3%)
Nodal marginal-zone lymphoma (NMZL)	27 (0.1%)	5 (0.0%)	<5	22 (0.3%)
Splenic marginal-zone lymphoma (SMZL)	143 (0.4%)	72 (0.2%)	<5	69 (1.1%)
Mantle-cell lymphoma	1,466 (3.9%)	1,116 (3.8%)	87 (4.4%)	263 (4.1%)
Myeloma and related plasma-cell disorders	8,953 (23.9%)	6,904 (23.7%)	491 (25.0%)	1,558 (24.4%)
Multiple myeloma (MM)	8,629 (23.0%)	6,673 (22.9%)	464 (23.6%)	1,492 (23.3%)
Plasma-cell leukaemia (PCL)	201 (0.5%)	146 (0.5%)	18 (0.9%)	37 (0.6%)
Solitary myeloma/plasmacytoma	123 (0.3%)	85 (0.3%)	9 (0.5%)	29 (0.5%)
T-cell non-Hodgkin lymphoma	1,616 (4.3%)	1,262 (4.3%)	74 (3.8%)	280 (4.4%)
Angioimmunoblastic T-cell lymphoma (AITL)	483 (1.3%)	351 (1.2%)	26 (1.3%)	106 (1.7%)
Anaplastic large-cell lymphoma (ALCL)	410 (1.1%)	339 (1.2%)	12 (0.6%)	59 (0.9%)
Peripheral T-cell lymphoma (PTCL)	522 (1.4%)	444 (1.5%)	17 (0.9%)	61 (1.0%)
T-cell lymphoma, unspecified (TCL)	201 (0.5%)	128 (0.4%)	19 (1.0%)	54 (0.8%)
Recorded performance status, *n* (%)				
Performance status 0	14,662 (53.8%)	11,842 (53.2%)	858 (54.3%)	1,962 (57.9%)
Performance status 1	8,262 (30.3%)	6,838 (30.7%)	434 (27.5%)	990 (29.2%)
Performance status 2	2,557 (9.4%)	2,077 (9.3%)	187 (11.8%)	293 (8.6%)
Performance status 3	1,273 (4.7%)	1,063 (4.8%)	88 (5.6%)	122 (3.6%)
Performance status 4	481 (1.8%)	444 (2.0%)	13 (0.8%)	24 (0.7%)
Missing performance status[a]	10,232 (27.3%)	6,844 (23.5%)	387 (19.7%)	3,001 (46.9%)
Disease-specific prognostic risk category, *n* (%)				
Low	3,989 (17.6%)	3,231 (17.4%)	243 (17.6%)	515 (18.9%)
Intermediate	10,513 (46.4%)	8,608 (46.4%)	639 (46.3%)	1,266 (46.5%)
High / very high[b]	8,161 (36.0%)	6,721 (36.2%)	497 (36.0%)	943 (34.6%)
Missing risk category[a]	14,804 (39.5%)	10,548 (36.2%)	588 (29.9%)	3,668 (57.4%)
Pre-treatment record summaries, median (IQR)[c]				
Recorded clinical-history span, years	**[Re-export]**	**[Re-export]**	**[Re-export]**	**[Re-export]**
Clinical-coded records	**[Re-export]**	**[Re-export]**	**[Re-export]**	**[Re-export]**
Distinct diagnosis codes	21.0 (13.0–32.0)	20.0 (12.0–30.0)	24.0 (15.0–37.0)	25.0 (15.0–38.0)
Distinct medication codes	21.0 (12.0–34.0)	18.0 (10.0–30.0)	32.0 (21.0–46.0)	34.0 (22.0–49.0)
Laboratory-coded records	197.0 (0.0–482.0)	118.0 (0.0–346.0)	483.0 (279.5–803.5)	531.0 (302.0–887.0)

One record per patient in the historical 37,467-person review export; these values are not the current 37,557-person first-line source or its supervised analysis subset. Disease groups are from `cohort_grouped`, not the separate, incomplete `Disease` field. Indented rows show fine cohorts within their grouped parent; one-to-one groups are shown once. Disease percentages use the full cohort/split, not the parent group. A dash denotes an absent row in the source export, not a confirmed zero. Detailed treatment categories and scoring systems are reported in the supplementary tables. [a]Performance status and disease-specific prognostic risk categories are sourced from quality registries, whose disease and calendar-period coverage is incomplete. Category percentages use patients with nonmissing values; missingness percentages use the full cohort/split. The later-period decline in availability must be re-estimated from the current release before interpretation. [b]High and very high are combined only for this descriptive overview. The very-high category occurs in CLL-IPI (114 patients overall); risk categories from different scoring systems are not equivalent clinical scales. Original categories and scoring-system identities are retained in the supplement. [c]MEDS-derived values are provisional pending re-export after excluding demographic tokens from clinical-history summaries. Laboratory entries count coded records, not unique tests or verified source measurements. Follow-up is observed duration, not a reverse-Kaplan–Meier estimate. IQR, interquartile range. Primary-only small-cell masking in the source exports is not disclosure clearance; the complete publication bundle requires institutional review.

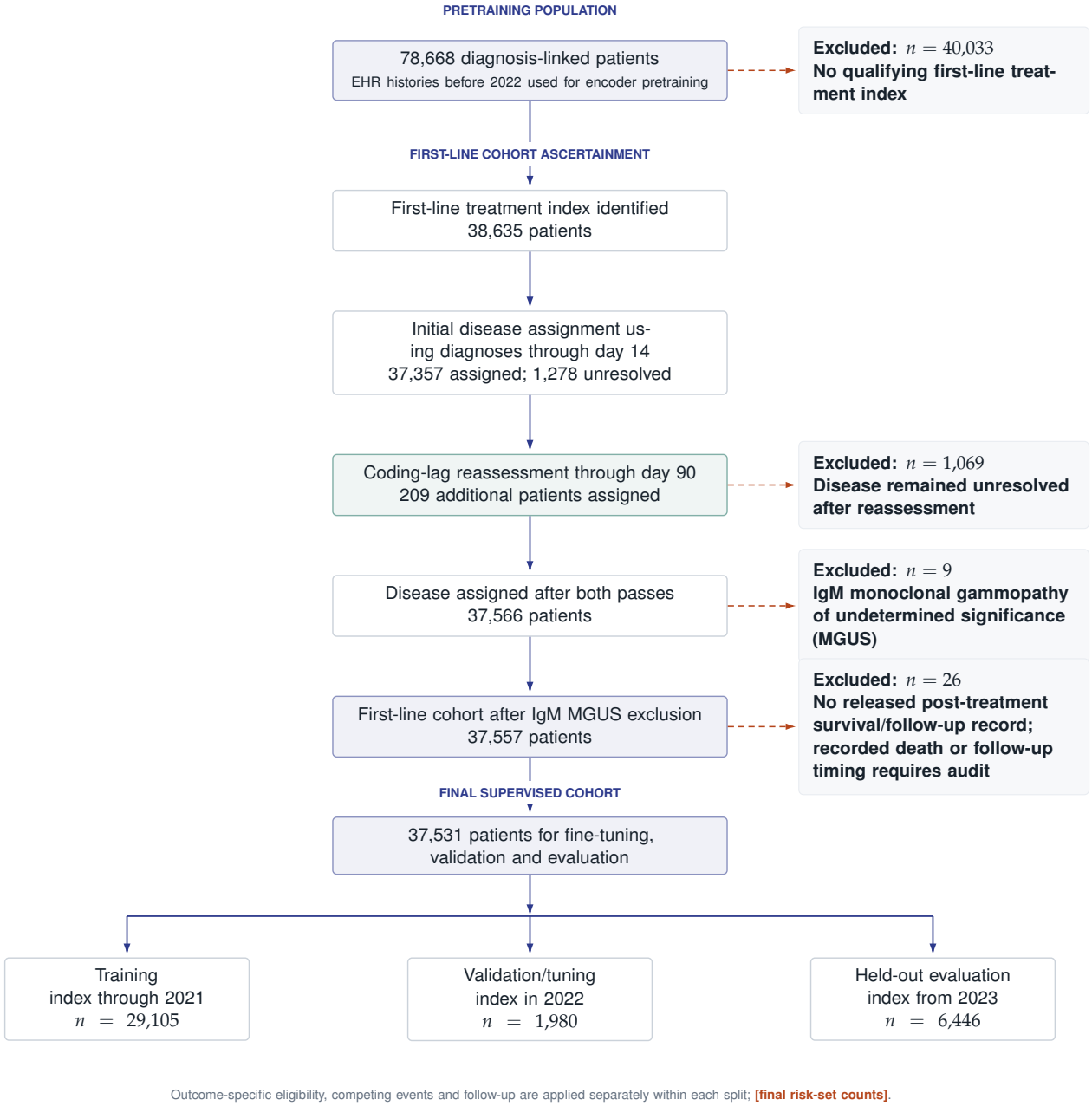


Figure 3. Construction of the longitudinal study population. The 78,668-patient diagnosis-linked population supplies pre-2022 histories for encoder pretraining; it is not patient-disjoint from later supervised splits. First-line ascertainment yields 37,557 patients, of whom 37,531 have released survival membership and a temporal supervision split. Initial disease assignment allows diagnoses through day 14 and coding-lag reassessment through day 90. The 26 patients without released supervised membership require source-level reconciliation. A sensitivity analysis will restrict routed disease labels to those available at treatment. Main-node counts are unique patients; outcome-specific risk sets are constructed after the temporal split and detailed attrition is in the supplement.

Figure 4 · Predictability atlas

Primary panel: held-out 365-day AUC for every outcome, ordered within clinical domains, with uncertainty and event support. Insets: outcome × cohort coverage and calibration. Supplement: the same atlas at all six horizons and within each fine cohort.

Figure 4. The landscape of predictable outcomes across haematology. Final artwork will be generated from the locked long-format results table; missing or unsupported cells remain visible as not estimable.

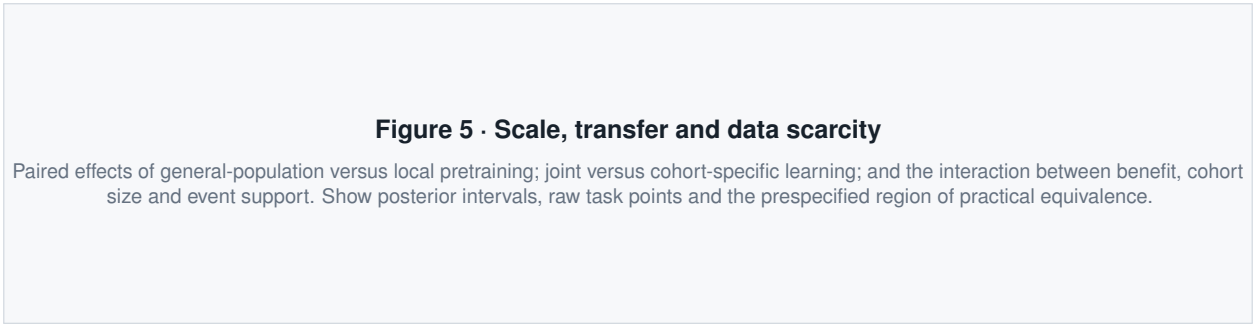


Figure 5. When broader pretraining and joint learning add value. The primary estimand is the paired held-out performance difference on identical patients, summarized hierarchically across tasks.

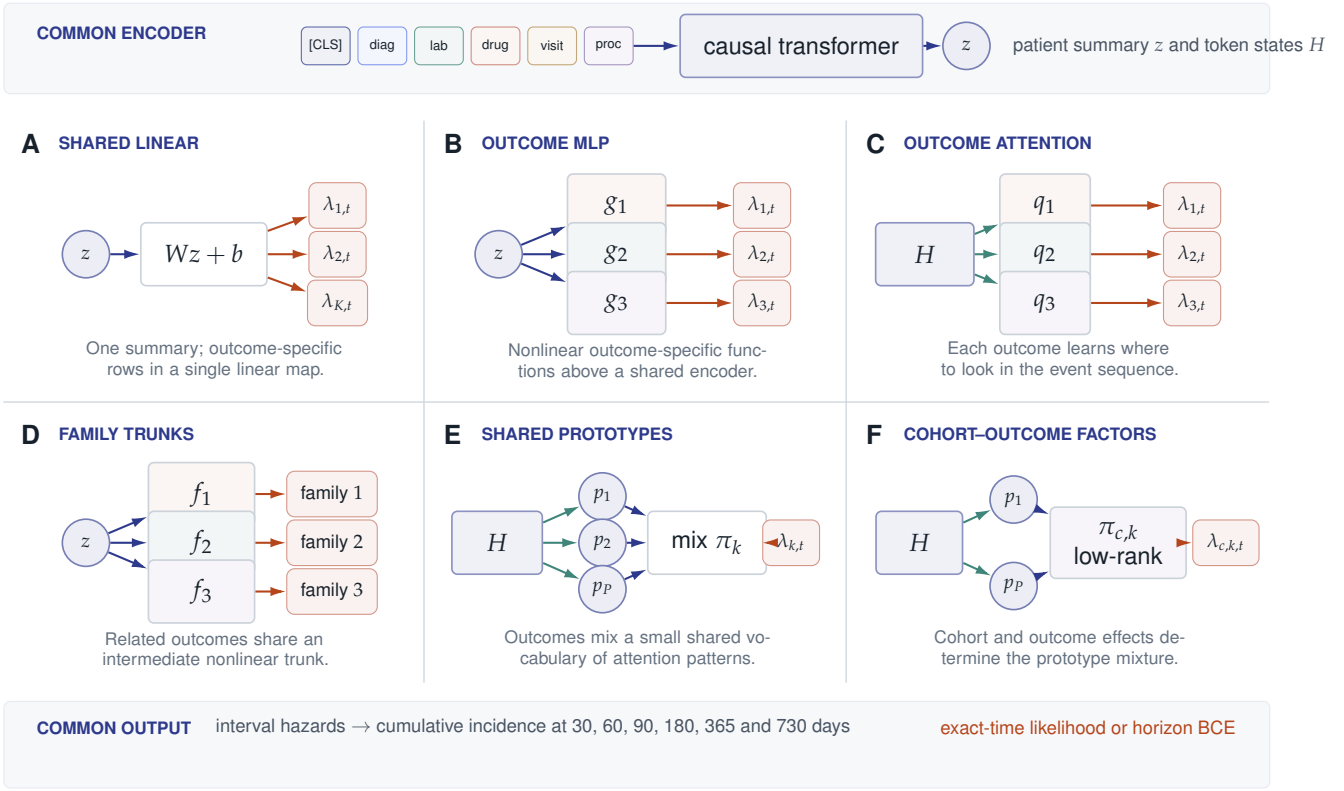


Figure 6. Multi-outcome readout architectures. Every model uses the same longitudinal encoder. **A**, a shared linear head predicts all outcome-specific interval hazards from one patient summary. **B**, an outcome MLP adds nonlinear task-specific capacity. **C**, learned outcome queries attend to different portions of the sequence. **D**, related outcomes share a nonlinear family trunk. **E**, outcomes mix a small set of shared sequence-attention prototypes. **F**, the factorized model combines global, cohort, outcome, and low-rank interaction effects to mix the same prototypes. Heads produce cause-specific interval hazards and cumulative incidence at all prespecified horizons; a multi-horizon binary cross-entropy objective is evaluated as an alternative to the exact-time likelihood.

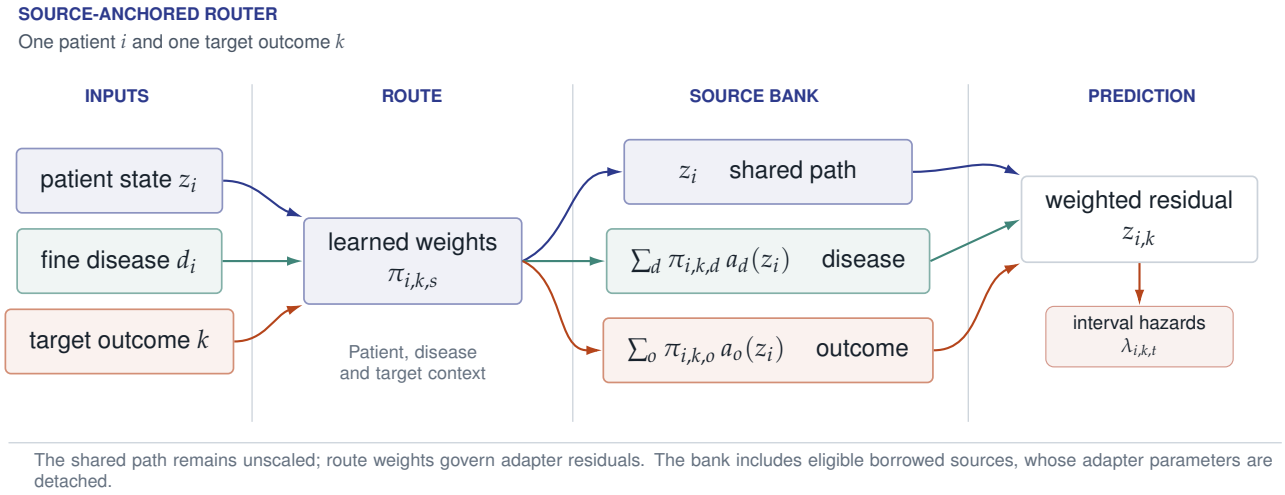


Figure 7. Source-anchored disease–outcome routing. For each patient and target outcome, the router mixes a shared representation with residual adapters indexed by fine disease and outcome. Its weights depend on the patient, disease and target; sources with no training support are masked and low-event sources are shrunk. An adapter is updated by its own eligible supervision, while borrowed adapter parameters are detached. The mixed representation enters a shared linear head for interval hazards and cumulative incidence. This is a model schematic, not an empirical route-weight or performance plot.

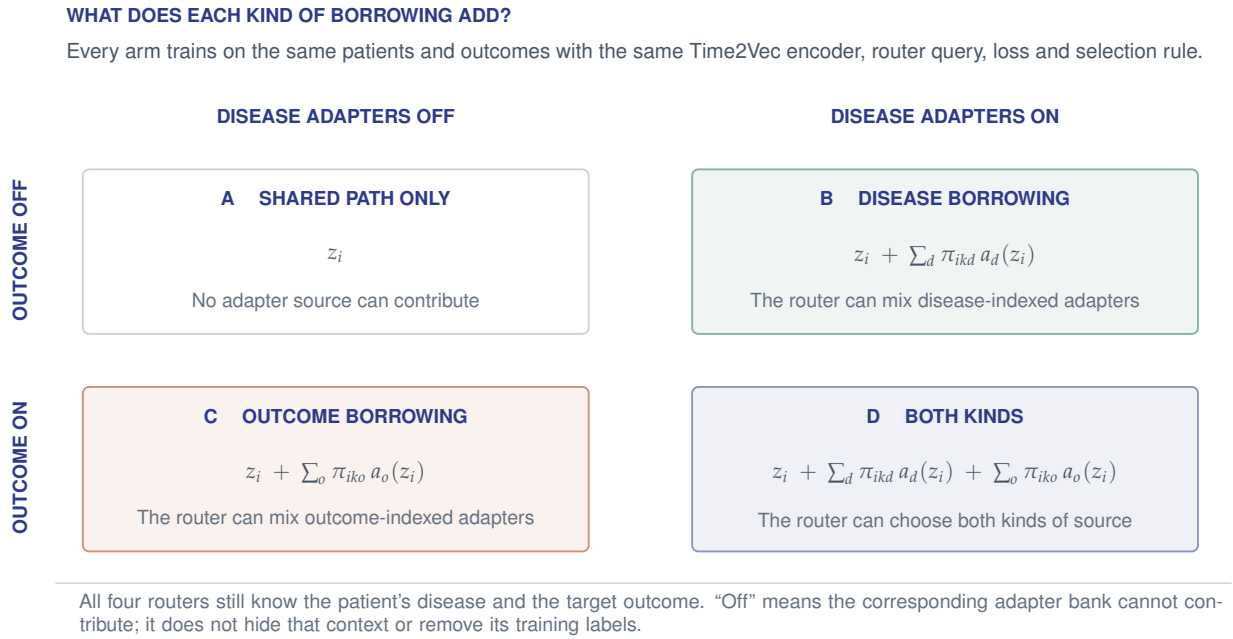


Figure 8. Two-by-two router source-type ablation. All arms retain the same Time2Vec initialization, pooled training patients and outcomes, interacting patient–disease–target query, shared path, hazard head, loss, and checkpoint rule. Only eligibility of disease-indexed and outcome-indexed adapter source banks changes. A-to-B and C-to-D estimate the conditional disease-source effect; A-to-C and B-to-D estimate the conditional outcome-source effect. The difference of these differences describes source-type interaction. No performance result is implied by panel placement or color.

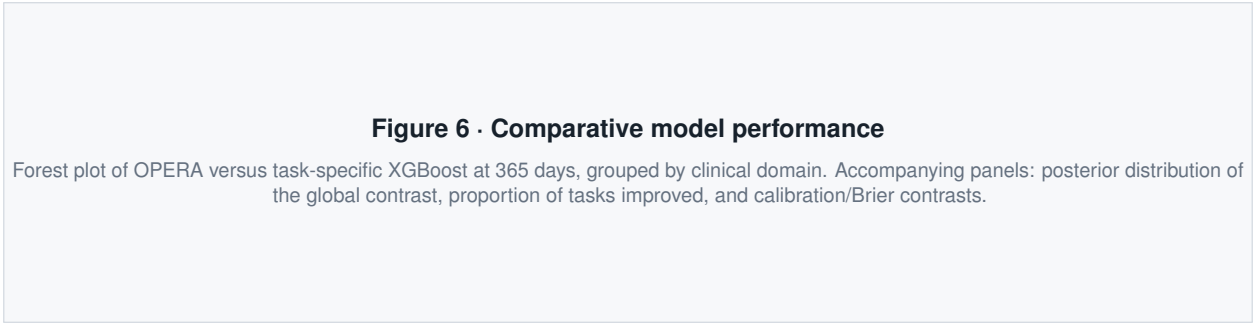


Figure 9. Sequence learning compared with task-specific tabular models. All contrasts use paired held-out predictions and identical endpoint eligibility.

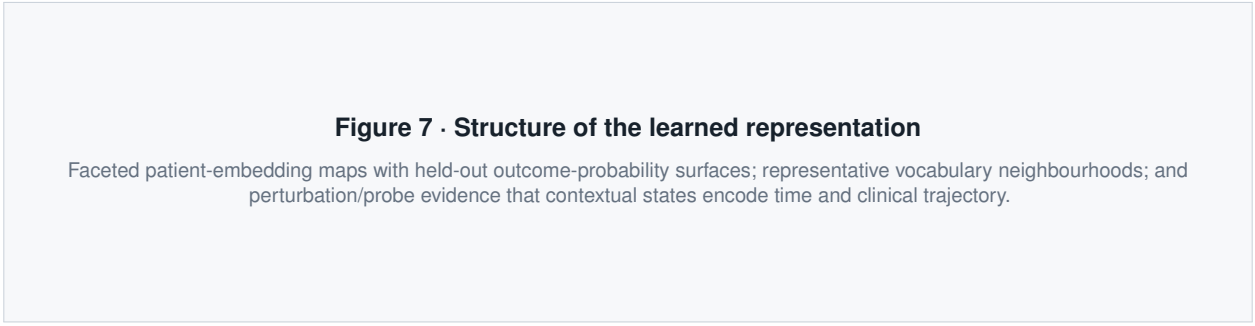


Figure 10. Clinical structure in the learned embedding space. Probability surfaces are descriptive held-out smooths, not independent estimates of model discrimination or causal effects.